Exploring a 2 by 2 Factorial with Control: A Case Study In SAS

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Introduction

Factorial experiments are widely used in scientific research for evaluating the effects of multiple factors simultaneously. The simplest form of the factorial design is a 2×2 factorial design which involves two factors. For each factor we have two levels, resulting in four treatment combinations. However, in many practical scenarios, researchers may wish to compare these treatments not only among themselves but also against a standard or baseline treatment referred to as a Control. In such cases, a 2×2 factorial design plus a Control is adopted. There is an important difference between the traditional factorial design with a Control and the problem which we attempt to address in this chapter. With the traditional factorial design with a Control, we see all possible combinations of the factor levels including the Control. For example, we may consider an experiment with two factors, type of fertilizer and irrigation each with two levels respectively. If A and B are the levels of fertilizer types and the levels of irrigation are Yes and No, where the latter is a control. Then for a traditional 2×2 factorial design, we would have all possible combinations: (A, Yes), (A, No), (B, Yes) and (B, No). Nevertheless, the type of problems that we try to address in this chapter involves a 2×2 factorial design where the control is serves as the fifth treatment.

In this chapter, we illustrate the concept of a 2×2 factorial design with control, where the control is considered as the fifth treatment using the example below. Researchers are interested in evaluating the effects of two drugs on blood pressure. They would like to determine if a Soluble or Non-soluble form makes a difference. Another factor that affects performance of the drug is dosage (10 units and 15 units). Because this is a new study, they also chose a Control group to serve as reference. In this experiment, the Control group refers to experimental units that did not receive any of the treatments.

A 2×2 factorial design with control may also be utilized in experiments where methods are being tested for the first time or in new areas of study. In the case where methods are being tested for the first time, researchers may be interested in comparing the main effects of treatments with Control. In the example we consider, we are interested in determining if the treatments are able to reduce blood pressure at all. In such instances, the Control group serves as a baseline and enables us to answer such a question. That is, if we find out that the blood pressure of individuals taking any of the drugs at different concentrations are on average lower than those in the Control group, then we know that taking a drug is at least better than taking nothing at all. Also, in situations where there may be a significant interaction effect, we can compare each treatment combination with the control.

Statistical Challenge

As we stated in the introduction, with a 2×2 factorial design with a Control where the Control appears in each possible treatment combination, the Control is considered as the fifth treatment and does not appear in any other treatment combination for the 2×2 factorial design plus Control. Some experiments include control and it is introduced as reference or baseline.

The statistical challenge is that, the control is not part of the factorial structure and thus, it creates a non-orthogonal or partially unbalanced design. This is a problem because the control does not contribute to the estimation of the main effects and interaction effects of factors. Nevertheless, the control shares the same experimental error term with the treatments. When the traditional 2×2 factorial design is used to analyze these kinds of experiments, the implication is that the control contributes to the marginal means and also, both main effects and interactions are interpretable and balanced.

The 2×2 factorial design with control enables the mean of the control to be estimated separately and computes factorial means only from the four factorial cells. Moreover, the variance of factorial effects are estimated only from factorial units and thus, control observations do not influence variance.

Exploratory Data Analysis

In our example which will be used to demonstrate the ideas in this chapter, we first conduct an exploratory data analysis to understand how data was generated, the type and description of the variables. We only present information that will be useful to the reader when using this design. We show the relative frequency density plot which gives us information about the distribution of the response. We also show the interaction plot and boxplot for all treatment combinations and the control.

Recall, the objective of the study is to evaluate the effects of two drugs on blood pressure. They would like to determine if a Soluble or Non-soluble form makes a difference. Another factor that affects performance of the drug is dosage (10 units and 15 units).

Data Description

Data was simulated from the normal distribution with mean of 0 and variance of 5. To set up the 2×2 factorial design with control in SAS, we first set up the treatment combination of drug type and dosage as a traditional factorial design and only add the control as an additional factor.

There were 15 observations with three replicates from each treatment combination and the control. Both drug type (Soluble and Non-soluble) and dosage (10 units and 15 units) were considered as qualitative variables. Thus, the reader can think of the levels of dosage as low and high for simplicity. The response variable in this experiment is blood pressure. Each individual in this experiment is considered as an experimental unit.

```
/* creates a data frame*/
Data bpdata;
/* variables or column names*/
INPUT drug$ dosage BloodPressure control$;
/* conditional statement identifying control and treatments*/
if drug='control' then do;
drug ='contrl';
forms = 0;
formn = 0;
end;
else if drug ='Soluble' then do;
forms = dosage;
formn = 0;
end;
else if drug='Non-sol' then do;
forms = 0;
formn = dosage;
end;

cards;
Soluble 10 128.1037837 n
Soluble 10 125.178207 n
Soluble 10 128.8657724 n
Non-sol 10 136.3624454 n
Non-sol 10 131.8548772 n
Non-sol 10 129.1857283 n
Soluble 15 127.9855594 n
Soluble 15 125.6000534 n
Soluble 15 124.2751741 n
Non-sol 15 136.5107568 n
Non-sol 15 131.021518 n
Non-sol 15 124.2247298 n
control 0 132.8951812 y
control 0 138.8245481 y
control 0 140.3308106 y

;
/*prints data*/
proc print;
run;
```

Obs	drug	dosage	BloodPressure	control	forms	formn
1	Soluble	10	128.104	n	10	0
2	Soluble	10	125.178	n	10	0
3	Soluble	10	128.866	n	10	0
4	Non-sol	10	136.362	n	0	10
5	Non-sol	10	131.855	n	0	10
6	Non-sol	10	129.186	n	0	10
7	Soluble	15	127.986	n	15	0
8	Soluble	15	125.600	n	15	0
9	Soluble	15	124.275	n	15	0
10	Non-sol	15	136.511	n	0	15
11	Non-sol	15	131.022	n	0	15
12	Non-sol	15	124.225	n	0	15
13	contrl	0	132.895	y	0	0
14	contrl	0	138.825	y	0	0
15	contrl	0	140.331	y	0	0

Figure 1: Data for 2 by 2 factorial with control

Figure 1 shows the relative frequency density plot which displays the distribution of blood pressure measurements in the data. The x-axis represents blood pressure values, while the y-axis shows relative frequency density, meaning the total area under the bars sums to one. Most observations are concentrated between approximately 124 and 136, indicating this range contains the highest relative density of measurements. The distribution shows moderate spread, with fewer observations at the higher end near 140 mmHg, suggesting slight right-skewness. No extreme outliers are apparent, although the density decreases toward the upper tail. Overall, the plot suggests that blood pressure values are clustered around the mid-120s to mid-130s, with relatively low variability outside this range.

Figure 2 displays the distribution of blood pressure across different drug types (Soluble, Non-soluble, and Control) at three dosage levels (0, 10, and 15 units). At dosage 0, the control group exhibits the highest median blood pressure, with values centered around the upper 130s, and relatively low variability. This suggests elevated blood pressure levels in the absence of treatment. No soluble or non-soluble drug groups are present at this dosage level.

At dosage 10, both soluble and non-soluble drugs show reduced median blood pressure compared to the control at dosage 0. The soluble drug group has the lowest median blood pressure and a relatively narrow interquartile range, indicating more consistent responses. In contrast, the non-soluble drug group shows a higher median and greater variability, suggesting less uniform treatment effects.

At dosage 15, the soluble drug continues to demonstrate the lowest median blood pressure with minimal variability, indicating a strong and consistent treatment effect. The non-soluble drug group again shows higher median values and a wider spread, with observations extending across a broader range, reflecting increased variability in response.

```
proc sgplot data=bpdata;
    vbox BloodPressure / category=drug group=dosage;
    yaxis grid;
run;
```

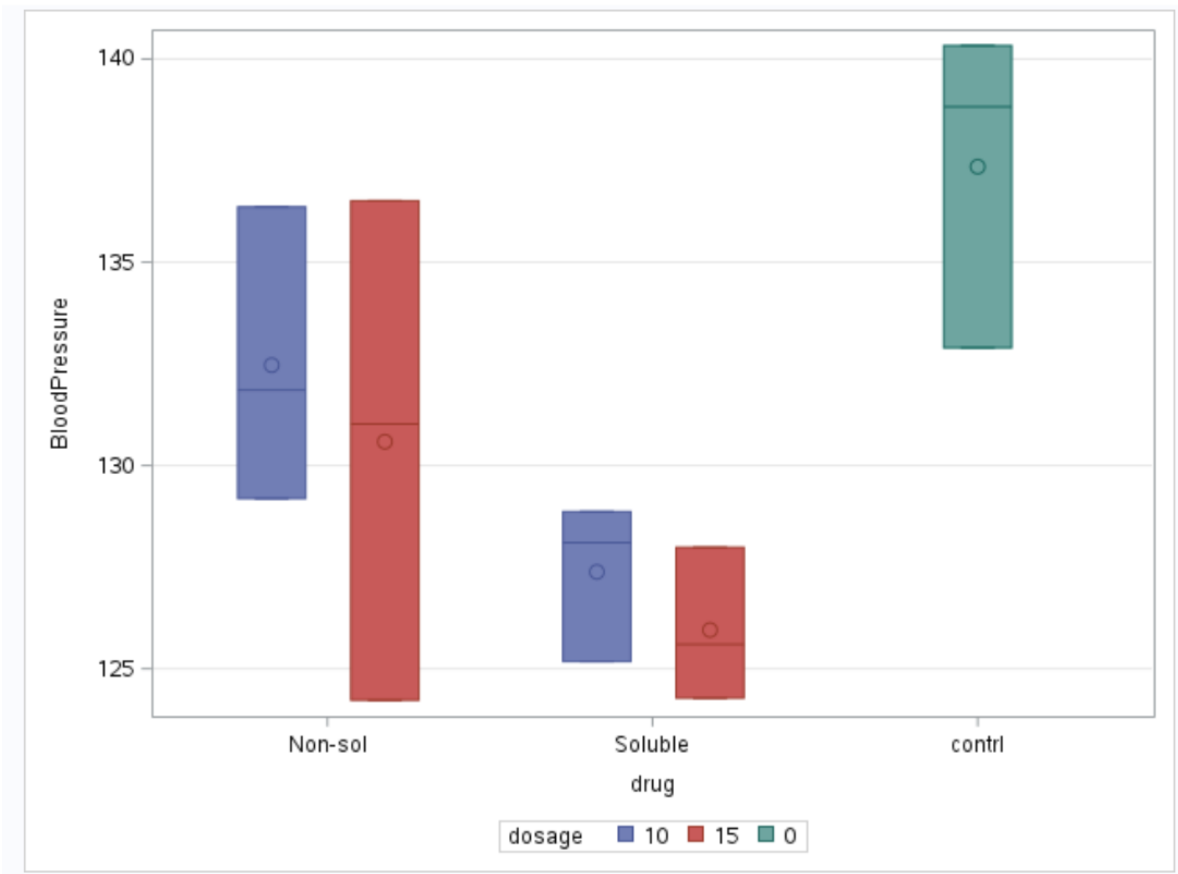

Figure 2: Boxplot of BloodPressure

Figure 3 shows the interaction plot for drug type and dosage. The absence of an interaction for a 2×2 factorial design is usually illustrated by parallel lines. From the interaction plot shown in Figure 3, we see parallel lines which shows there is no interaction between the two factors.

This implies that the effect of one factor, say drug type on blood pressure, is the same at all levels of dosage. Thus, the two factors are independent of each other. Nevertheless, we need to look at the anova table.

```
data bp_plot;
    set bpdata;
    if drug ^= "contrl";
run;

/* Compute mean BloodPressure for Drug × Dosage */
proc means data=bp_plot noprint;
    class drug dosage;
    var BloodPressure;
    output out=means_plot mean=MeanBP;
run;

/* Remove _TYPE_ and _FREQ_ rows from PROC MEANS */
data means_plot_clean;
    set means_plot;
    if _TYPE_ = 3; /* Keeps only Drug × Dosage combos */
run;

/* Get Control mean */
proc means data=bpdata noprint;
    where drug = "contrl";
    var BloodPressure;
    output out=control_mean mean=ControlMean;
run;

/* Add ControlMean to all rows for plotting */
data plot_ready;
    if _n_ = 1 then set control_mean;
    set means_plot_clean;
run;

/* Interaction Plot with Control Mean Line */
proc sgplot data=plot_ready;
```

```
    series x=drug y=MeanBP
      / group=dosage lineattrs=(thickness=2) markers;
    refline ControlMean
      / axis=y lineattrs=(color=darkgreen pattern=shortdash thickness=2)
      label="Control Mean";
    yaxis label="Average Blood Pressure";
    xaxis label="drug";
    keylegend / location=inside position=topright across=1;
    title "Interaction Plot of Drug and Dosage (with Control Reference)";
run;
```

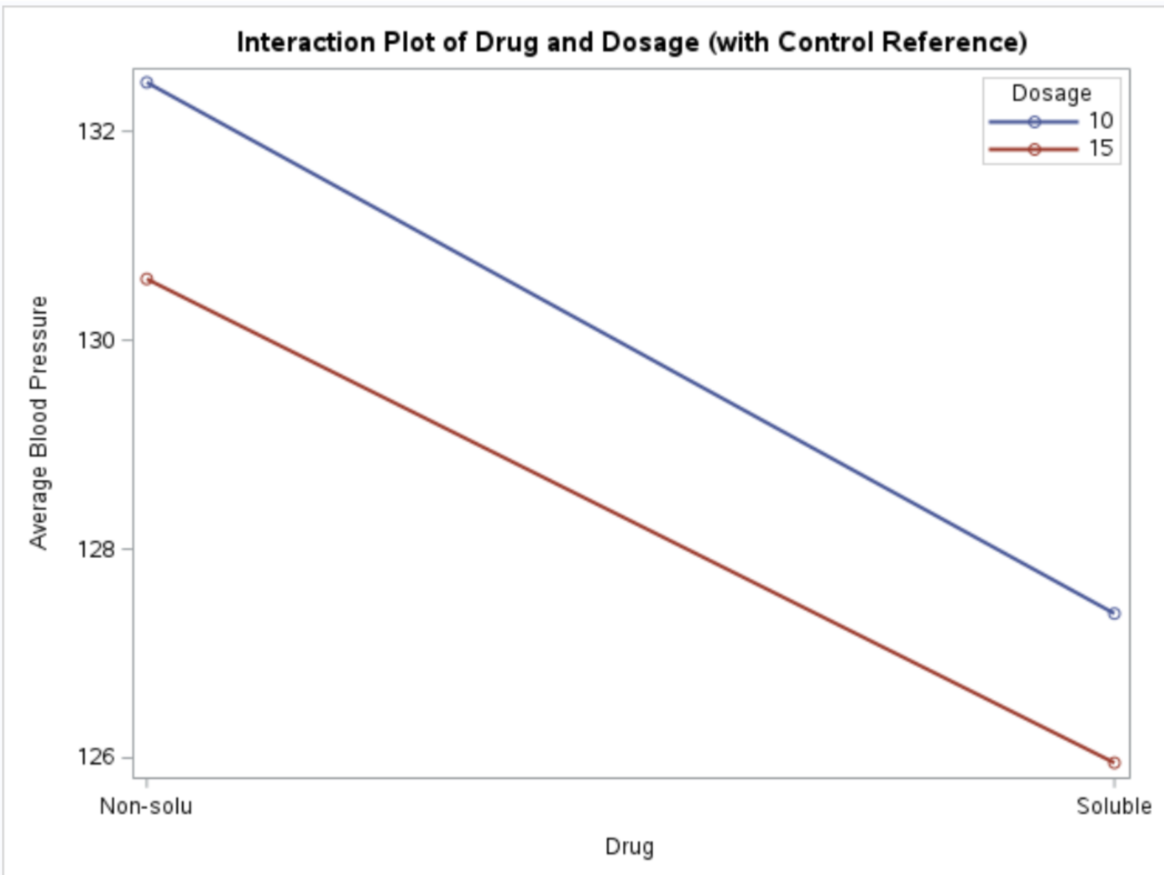

Figure 3: Interaction Plot of Drug and Dosage with Control as reference

Advanced Analysis

Recall, the main objective of the experiment is to understand how drug type and dosage levels affect blood pressure and also how the treatments perform in comparison to the control. Thus, we seek to answer two important questions with a single model.

The statistical model thus seek to evaluate whether average blood pressure of individuals taking a soluble drug differs from the blood pressure of individuals with non-soluble drug. Moreover, the model enables us to compare average blood pressure when a treatment was administered to that of the control group who did not receive any treatment all.

Similarly, we are able to evaluate the effect of the dosage levels and compare their effect on blood pressure. In this chapter, we consider a simple linear model with assumption that blood pressure is independent and identically distributed and that we have homogeneous variance.

Statistical Model

The linear model for a factorial design with control is given as;

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \gamma I_k + \varepsilon_{ijk}$$

and,

$$I_k = \begin{cases} 1, & \text{if the observation belongs to the control treatment,} \\ 0, & \text{if the observation belongs to one of the factorial treatments.} \end{cases}$$

$i = 1, 2,\ j = 1, 2, 3$ and $k = 0, 1$

where:

y_{ijk} is the blood pressure of the lth individual using the ith type of drug at the jth dose

μ represents the baseline mean of the factorial treatments

α_i is the effects of the ith drug type

β_j is the effects of the jth dosage

$(\alpha\beta)_{ij}$ is their interaction effect

γ measures the difference between the control treatment and the factorial baseline

ε_{ijk} is assumed to be independently and normally distributed with mean zero and constant variance σ^2.

Table 3 shows the ANOVA table when the data is analyzed as a traditional 2 by 2 factorial design. Note that, analyzing the experiment in this manner is incorrect. Looking at the Figure 4, we see that there is a significant effect of dosage with p-value = 0.02112. This is because the control is embedded in dosage and thus the estimated mean for dosage includes that of control. With this result, a researcher may believe that dosage of the drug was significant and may continue to explore which dosage led to the highest effect.

```
/* Traditional factorial design */
proc glimmix data = bpdata1;
class drug dosage;
model BloodPressure = drug dosage
  drug*dosage / solution ddfm=kr;
lsmeans drug dosage drug*dosage;
run;
```

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
drug	1	10	4.80	0.0533
dosage	1	10	0.56	0.4727
drug*dosage	1	10	0.01	0.9207

Figure 4: ANOVA for traditional 2 by 2 factorial design

Table 4 takes a look at the correct model. From Table 4, we see that the p-value for the control shown in the anova table is statistically significant, $p - value = 0.0076$ and the p-value for dosage by drug interaction term is 0.9207. These values correspond to the interaction plot we saw earlier. Since there is no significant interaction effect, we can look at the main effects for Drug (soluble or non-soluble) to determine if there is any significant difference.

```
proc glimmix data = bpdata;
class drug dosage control;
model BloodPressure =  control drug(control) dosage(control)
  drug*dosage(control)/solution ddfm=kr;
lsmeans control drug(control) dosage(control);
run;
```

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
control	1	10	11.07	0.0076
drug(control)	1	10	4.80	0.0533
dosage(control)	1	10	0.56	0.4727
drug*dosage(control)	1	10	0.01	0.9207

Figure 5: ANOVA for corrected 2 by 2 factorial design

We proceed with further analysis by looking at the main and simple effects of only the correct model. Table 5 shows the estimated marginal means for drug type. For drug type soluble and non-soluble, the estimated marginal means are 126.67 and 131.53 respectively with standard error 1.5686. The estimated mean for the control group is also 137 with standard deviation of 2.22. The 95 percent confidence intervals are also shown in the last two columns of the same table.

```
 ## lsmeans statement to obtain estimated marginal means
lsmeans drug(control) / diff= control adjust=dunnett cl;
run;
```

drug(control) Least Squares Means									
drug	control	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
Non-sol	n	131.53	1.5686	10	83.85	<.0001	0.05	128.03	135.02
Soluble	n	126.67	1.5686	10	80.75	<.0001	0.05	123.17	130.16
contrl	y	137.35	2.2183	10	61.92	<.0001	0.05	132.41	142.29

Figure 6: Estimated marginal means

Table 6 shows the estimated marginal means for dosage. When dosage levels are at 10 and 15 units, the estimated marginal means are 130 and 128 respectively with standard error 1.57. The estimated mean for the control group is again 137 with standard deviation of 2.22 indicating that the control group did not receive any treatment.

```
 ## lsmeans statement to obtain estimated marginal means
lsmeans dosage(control) / diff= control adjust=dunnett cl;
run;
```

dosage(control) Least Squares Means						
control	dosage	Estimate	Standard Error	DF	t Value	Pr > |t|
n	10	129.93	1.5686	10	82.83	<.0001
n	15	128.27	1.5686	10	81.77	<.0001
y	0	137.35	2.2183	10	61.92	<.0001

Figure 7: Estimated marginal means

Simple Effects

Next, we show the simple effects of drug and dosage. A simple effect refers to the effect of one treatment factor at a specific level of another factor. One typically would need to look at simple effects if there is a significant interaction effect. It is used in factorial experiments to understand how factors interact. In our example, there was no significant interaction effect but we show the results here for completeness.

From Table 7, the simple effect for dosage at 10 and 15 units when a soluble drug is used are 127 and 126 respectively. For non-soluble drug, the simple effect of dosage at 10 and 15 units are 132 and 131 respectively. Looking at the simple effects, we see a lower blood pressure when the drug is soluble. Nevertheless, without any drug or dosage, the blood pressure is high.

```
 ## lsmeans statement to obtain estimated marginal means
lsmeans dosage*drug(control) / diff= control adjust=dunnett cl;
run;
```

drug*dosage(control) Least Squares Means										
drug	control	dosage	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
Non-sol	n	10	132.47	2.2183	10	59.72	<.0001	0.05	127.52	137.41
Non-sol	n	15	130.59	2.2183	10	58.87	<.0001	0.05	125.64	135.53
Soluble	n	10	127.38	2.2183	10	57.42	<.0001	0.05	122.44	132.33
Soluble	n	15	125.95	2.2183	10	56.78	<.0001	0.05	121.01	130.90
contrl	y	0	137.35	2.2183	10	61.92	<.0001	0.05	132.41	142.29

Figure 8: Simple effects

The same results from Table 7 are obtained when we find the simple effect of drug type at given dosage levels. At dosage level of 10 units, the simple effect for soluble and non-soluble drugs are 127 and 132 respectively. While at a dosage level of 15 units, the simple effects of soluble and non-soluble drugs are 126 and 131 respectively.

Recall, the 2×2 factorial design with control enables us to also compare the main effect of treatments to the control group. This is possible because of the variable Control in the last column of the data set. It serves as an indicator variable and thus enables the model to average over the values of the treatments and compare them to the control group. From Table 9, we see that on average the estimated blood pressure for individuals receiving a treatment compared to control group was 129 and 137 respectively. Thus, the blood pressure for individuals in the control group is higher on average compared to individuals who receive some type of treatment. The difference between the two values is shown by the contrast statement. From Table 9, we see that the difference is 8.25 and with a standard deviation of 2.48. The p-value is 0.0076, showing that there is a statistically significant difference between individuals that received some sort of treatment and the control group at $\alpha = 0.05$.

```
 ## lsmeans statement to obtain estimated marginal means
lsmeans control / diff= control adjust=dunnett cl;
run;
```

control Least Squares Means								
control	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
n	129.10	1.1092	10	116.39	<.0001	0.05	126.63	131.57
y	137.35	2.2183	10	61.92	<.0001	0.05	132.41	142.29

Figure 9: Simple effects

```
 ## lsmeans statement to obtain estimated marginal means
lsmeans control / diff= control adjust=dunnett cl;
run;
```

Differences of control Least Squares Means Adjustment for Multiple Comparisons: Dunnett												
control	_control	Estimate	Standard Error	DF	t Value	Pr > |t|	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper
y	n	8.2528	2.4802	10	3.33	0.0076	0.0076	0.05	2.7266	13.7789	2.7266	13.7790

Figure 10: Difference between treatment and control

Finally, Table 10 shows differences between pairs of treatment. This is obtained by writing contrast statements. Writing contrasts is very useful especially when one has so many treatments and may only be interested in a few comparisons. Therefore, we would write some contrast to determine if there is a significant difference between soluble and non-soluble drugs, soluble and control, non-soluble and control. We used the dunnett adjustment to control for family-wise error.

From Table 10, there is no significant difference in effect between soluble and non-soluble drug and between non-soluble drug and control, p-values are 0.1314 and 0.1414 respectively at 5 percent significance level. Nevertheless, there is a significant difference in effect between soluble drug and the control group, p-value = 0.0076.

```
 ## lsmeans statement to obtain estimated marginal means
lsmeans dosage(control) / diff= control adjust=dunnett cl;
run;
```

Differences of drug(control) Least Squares Means Adjustment for Multiple Comparisons: Dunnett														
drug	control	_drug	_control	Estimate	Standard Error	DF	t Value	Pr > |t|	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper
Soluble	n	Non-sol	n	-4.8586	2.2183	10	-2.19	0.0533	0.0959	0.05	-9.8013	0.08416	-10.5882	0.8711
contrl	y	Non-sol	n	5.8235	2.7169	10	2.14	0.0577	0.1035	0.05	-0.2301	11.8771	-1.1939	12.8409

Figure 11: Difference between treatment and control

Conclusion

In this chapter, we considered the 2×2 factorial design with control and how experiments of this nature can be analyzed. The example used to illustrate the concepts in this chapter was about evaluating the effects of two drugs on blood pressure which consisted of two factors and a control group. This design allowed researchers to answer some important questions which are usually present in studies that may be groundbreaking or there may not be enough research already done in that field.

The analysis showed that, without the application of any treatment, blood pressure among individuals may stay high in comparison to when a treatment is administered. Moreover, the type of the drug, whether soluble or not may have an effect on blood pressure. Nevertheless, there was no statistical evidence to support the effect caused by an increase in dosage. Also, for the example and data set used for analysis, there is no interaction between the two factors. Though in this illustration we simulated our data, the concepts are applicable to several studies of this nature. We also want to highlight how analyzing experiments of this nature as a traditional factorial design can change the inference space of the study.